SAMUEL FAN / PRODUCT & INTERACTION DESIGN

Sustained product work.
From requirements to delivery.

PRIMARY PROJECT / MY LONGEST INVOLVEMENT

Fire Equipment Management.

Requirements · Cross-app interaction
Test/build handoff · Ongoing iteration

Company project / 2026-Present

Site + service needs
Owner and premises context

Field work
Installation and inspection journeys

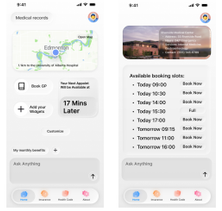
Review + next action
Progress across roles

SUPPORTING PROJECTS



Smart Livestock Monitoring

Field research · UI · Android MVP



One System, One Journey

Figma interface design · Academic team concept

Sydney, Australia · fanxin0528@gmail.com · 0416358898

01 / PRIMARY PROJECT · LONGEST INVOLVEMENT

Making field work connect across roles.

Fire Equipment Management Platform

CONTEXT	STAGE	MY FOCUS
Company project · 2026	Delivery and ongoing iteration	Requirements · Interaction direction · Handoff

My longest project involvement connects requirements, cross-app interaction, testing and delivery. The central design question stayed consistent: how does each role know what to do next?

The challenge

A site owner, field inspector, installation team and administrator see different parts of the same work. An unclear entry point, mismatched task status or ambiguous completion step can interrupt the handoff between them.

The design problem extends beyond individual screens: it includes the business object being managed, the person responsible and the evidence of the next action.

My contribution

I prepared overseas requirements discussions, clarified site and role boundaries, reviewed cross-app interactions and helped turn observed issues into prioritised fixes and checks.

I worked with product, backend and hardware colleagues. Engineering teammates and AI-assisted workflows supported implementation while I focused on requirements, interaction review and delivery coordination.

Four surfaces, one connected service

01	02	03	04
Site owner	**Field inspector**	**Installation team**	**Administrator**
Manage sites	Receive assignments	Install equipment	Coordinate work
Inspect equipment	Check evidence	Review status	Review certificates
Request support	Submit conclusions	Handle field work	Handle alert cases

Anonymised role overview, reconstructed for this case study.

Delivery stage. The work progressed through Android build handoffs and continued workflow refinement.

02 / REQUIREMENTS

Start with the right unit of work.

Stakeholder clarification shaped the site model and the division of responsibilities.

What needed clarification

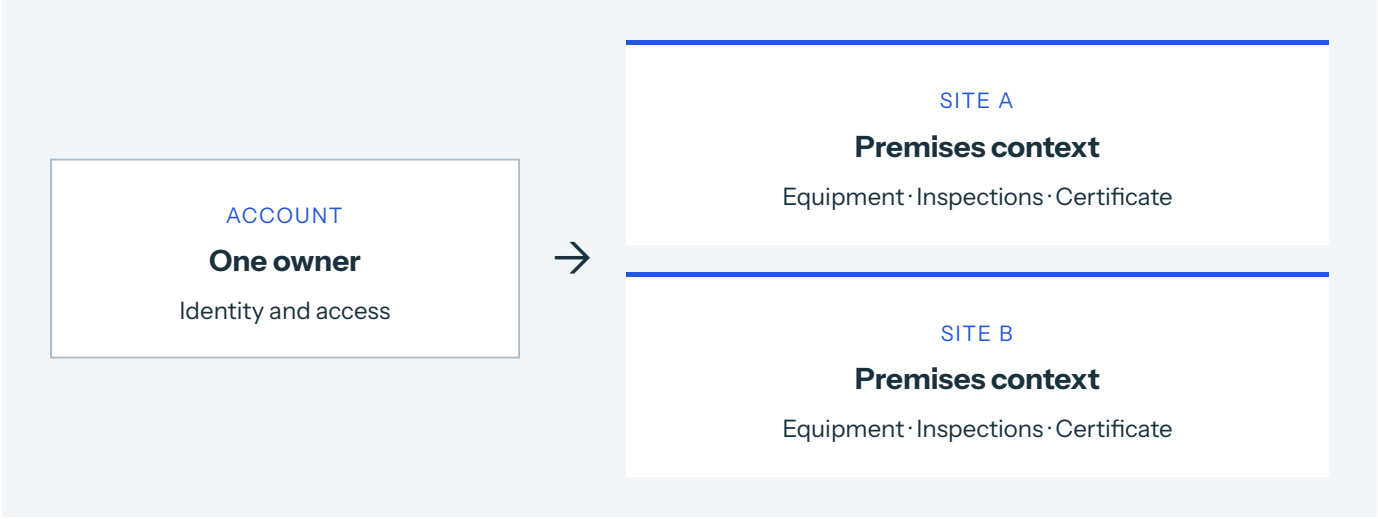
Before an overseas discussion, I identified questions about what a certificate covered, how related charges were separated, who installed smoke detectors and which interface should hold those responsibilities.

I requested an English discussion guide, then supplied the meeting transcript for follow-up synthesis. The stakeholder input helped distinguish site certification from the connected-device service.

What I took forward

I proposed allowing an owner to manage multiple sites. Later, I clarified that a site's personal or organisational classification should not be duplicated as an account type during registration.

This kept two concepts separate: who signs in and which places they manage. Installation responsibilities also needed their own operational surface rather than being placed indiscriminately in the owner or inspector app.



Reconstructed conceptual model with fictional sites.

THE PRODUCT DECISION

Put each choice where it belongs.

Account registration should establish the person. Site management should establish the premises. Installation and inspection should follow their actual responsibilities.

03 / BUSINESS JOURNEY

Connect the work, then follow it through.

A broad service journey across site ownership, field work and administration.

I reviewed the platform as connected business journeys: establishing a site, coordinating work, and making the outcome visible to the next role.

ESTABLISH THE CONTEXT

Owner + site

Identify the premises and its equipment; understand the certificate or installation need.

↓

COORDINATE FIELD WORK

Installation

Request support, assign the relevant team and review installation progress.

Inspection

Assign a check, accept the task, complete field work and submit the conclusion.

↓

REVIEW + CONTINUE

Administrator + owner

Review the outcome, make the next status clear and continue with certification or follow-up work.

Anonymised business-level reconstruction. The installation and inspection paths show different roles, not one mandatory sequence.

Where I focused

I clarified what belonged to the account or site, questioned unreachable request paths, and reviewed whether a task could continue naturally after acceptance. I also reported when a completed field task still appeared open elsewhere.

A separate urgent journey

Routine inspection was not the same job as emergency response. Later requirements brought owner reports and device alerts into an admin coordination workflow, with location, contact and explicit incident-end feedback.

> **The product lesson.** A completed action on one screen is a handoff to another person. I used those handoffs to frame review, prioritisation and the next check with the team.

04 / INTERACTION DIRECTION

Keep the task moving.

Reviewing hierarchy, status and transitions across the mobile apps and admin console.

I tested and reviewed the interfaces as connected tasks, asking what an operator should understand, choose and see after each action.

01 Show the problem, not just a category.

I asked for specific status wording rather than a generic warning or non-compliance label, supported by a consistent severity colour. I also reviewed drawer details and redundant close controls in the admin interface.

Illustrative copy direction: ~~Warning~~ → **Expires within 30 days**

02 Accept a task, then continue doing it.

On the inspector app, I wanted the assignment itself to have visual priority. After acceptance, the flow should continue into inspection instead of bouncing the person back to the list and forcing another search.

Read assignment → Accept task → Continue inspection

03 Use a shared interaction language.

I requested alignment between the installation app and the existing owner/inspector apps, including navigation, refresh behaviour, input and feedback. Earlier testing identified missing back navigation, unclear completion feedback and card layouts with weak information hierarchy.

Make handoff understandable

I requested separate usage guides for all four surfaces, written around human actions rather than internal technical calls. I also asked for step-by-step test instructions with an explicit result record.

Work with engineering

I provided interaction direction, concrete observations, sequencing and review. Engineering and AI-assisted workflows translated these into shared specifications, implementation and checks.

Copy and task-flow examples reconstructed from the review direction.

05 / WORKFLOW REFINEMENT

An alert needs a clear next action.

Moving from signal presentation to an understandable dispatch workspace.

Revisit who actually responds

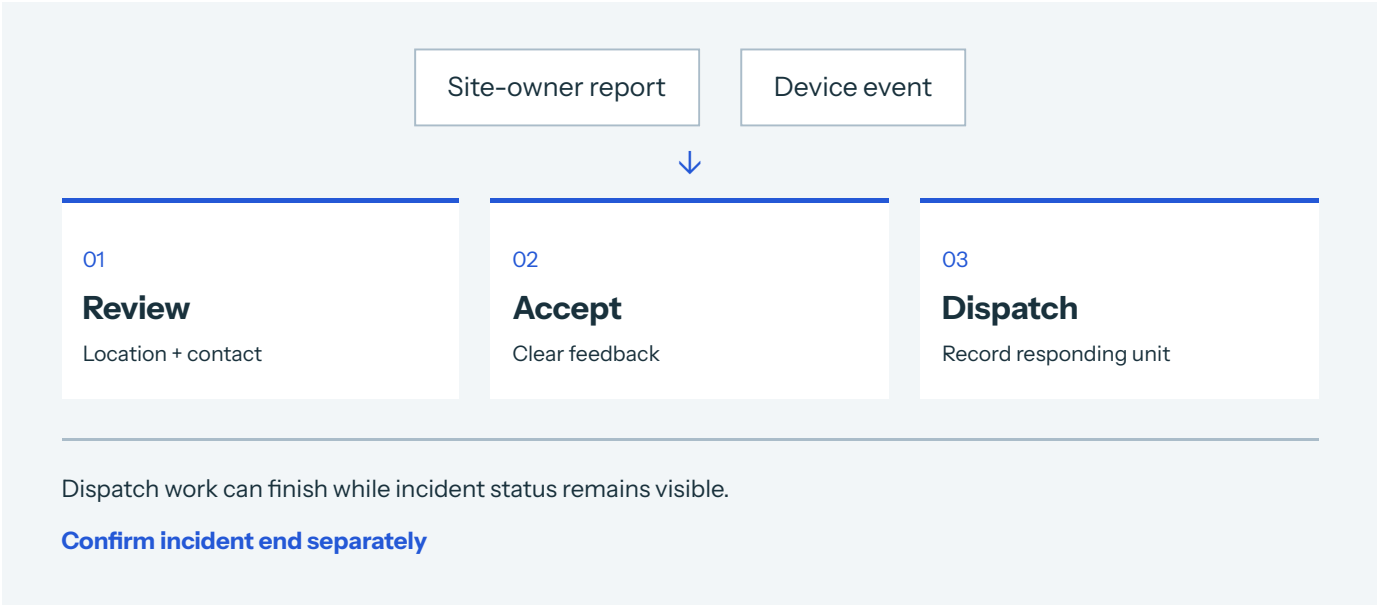
I raised the need for a site owner to report a fire and clarified that the inspector app was used for inspection staff, not every emergency responder. This moved emergency coordination towards the admin workspace.

I discussed location confirmation, accidental activation, responsible operators and contact information, while reviewing AI-assisted proposals for the case workflow and shared editing.

Let testing change the scope

After trying the flow, I found acceptance and completion unclear. I questioned the long reporting sequence because we had not yet established the real dispatcher's reporting needs.

I proposed a smaller first workflow centred on confirming the responding unit. I then asked how a dispatched alert would leave the map, exposing the difference between sending help and confirming that an incident had ended.



Anonymised workflow reconstruction based on the refined requirements.

The resulting direction

The subsequent team specification separated the dispatch queue from confirmation that an incident had ended. It retained site contacts, visible progress and accountable actions while reducing routine form filling. Device state and administrative case status remained distinct.

> **My role in the decision.** I challenged confusing transitions, narrowed the first scope and reviewed operational boundaries. AI and engineering contributed detailed case-state, editing and implementation solutions.

06 / ONGOING INVOLVEMENT

A project that kept evolving.

Continued involvement across changing requirements, build handoffs and operator feedback.

JUN	Clarify certificate scope, installation responsibilities and multi-site needs.
JUL	Review mobile consistency, operator guidance and end-to-end test steps.
AUG	Prioritise the smallest complete workflows and review cross-role interactions.
SEP	Coordinate a demonstration baseline; refine reporting, dispatch and subsequent workflows.

What was delivered

The project record includes signed Android build handoffs, role-specific usage guides, test instructions and later workflow revisions. Together, these artifacts connect product decisions with implementation and operator handoff.

Before a planned demonstration, I requested a return to a known-working build while keeping newer changes available for later iteration. This kept demonstration readiness separate from subsequent product iteration.

Keep each stage visible

Release records distinguish active builds, inactive candidates and demonstration packages. Hardware checks and human acceptance were tracked separately from automated or simulated verification.

This distinction guided the next step: a build can be ready for a scoped check while the wider service continues through hardware testing, operator feedback and acceptance.

WHAT THIS WORK SHOWS

Product judgement across the handoffs.

I can turn stakeholder ambiguity into concrete questions, keep role boundaries understandable, identify interaction breaks and coordinate the next verifiable slice with technical colleagues.

What I would carry into the next iteration

Observe the actual dispatcher and installer workflows, validate how much information is needed at each decision, then compare field findings with the current interaction sequence. Confirm real-device behaviour and acceptance independently of a build's machine checks.

Ongoing company project. Diagrams are anonymised reconstructions; confidential commercial details are omitted.

FIELD RESEARCH / PRODUCT DECISIONS / 2026

From the field to a testable direction.



Original field photograph / no people or identifying signage

I conducted three days of field research, then contributed UI, handoff and AI-assisted prototype iteration. The complete user flows, IA and specifications were team deliverables.



Actual Android MVP
Confirmed synthetic demo records

DECISIONS THAT SHAPED THE WORK

Different roles. Different evidence.

RESEARCH SYNTHESIS / REDRAWN FOR THIS CASE

WORKERS

Lower-text input

Graphical cues, scanning, photos

MANAGERS

Useful operating records

Fit existing routines

BANK MONITORING

Identity and data trust

Compare evidence and failure risks

INPUT CONSTRAINT

Some workers faced text-input friction. I brought lower-text approaches into the requirements discussion.

TECHNICAL TRADE-OFF

A shared link can create a blind spot. Compare local collection with GPS/BeiDou fallback by context.

PROTOTYPE SCOPE

I chose Flutter for the April MVP and prioritised cattle, alerts and weight entry with simulated data.

CHECKED IN REVIEW

Navigation visibility, alignment and consistency became concrete iteration items.

NEXT EVALUATION

Test core tasks with each role in context; integrate hardware before claiming field outcomes.

ACADEMIC TEAM WORK / 2025

DECO2200 / INTERACTION DESIGN STUDIO / 2025 S2

One System, One Journey.

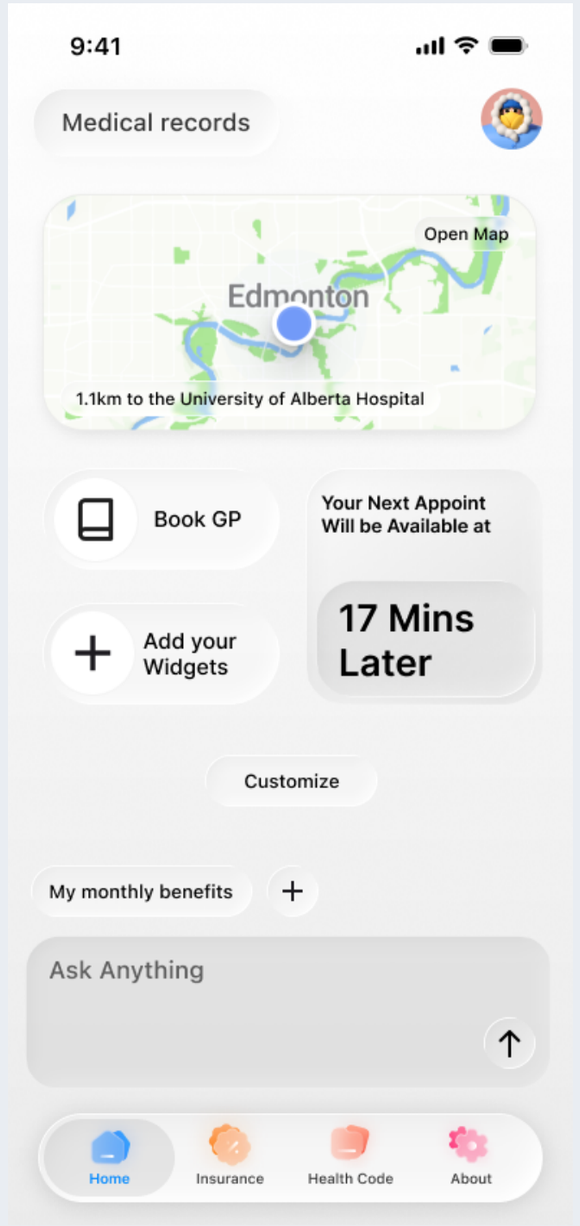
An omni-channel healthcare concept, shown through the team's original Figma screens.

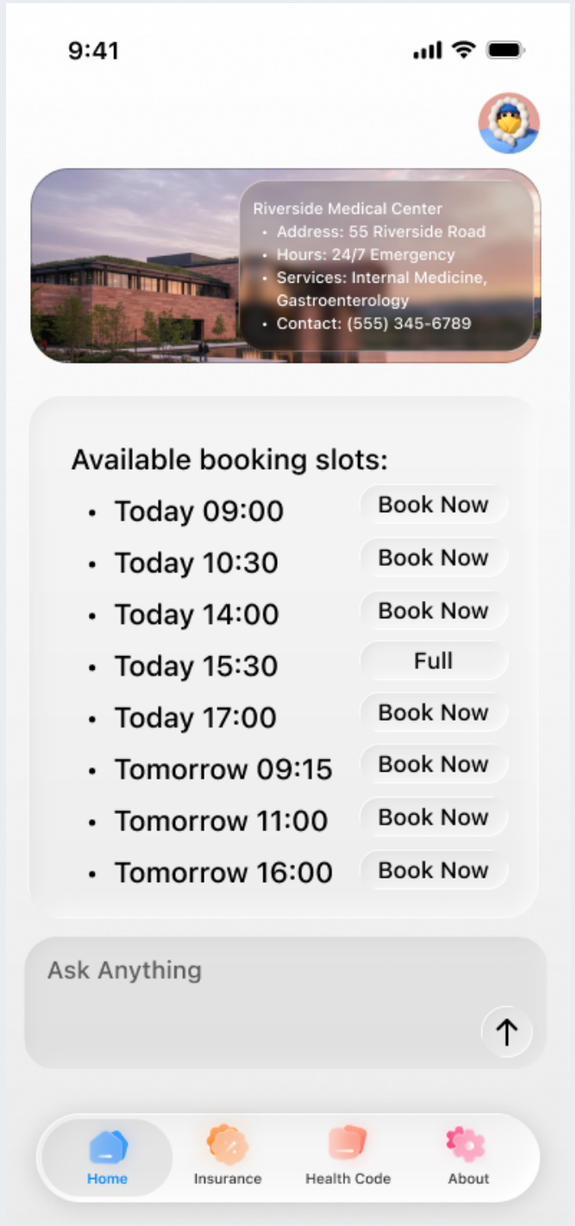
VISUAL STUDY

Soft surfaces, rounded panels and colourful icon accents organise the mobile experience.

MY CONTRIBUTION

I created the Figma interface design for this academic team project. Research and other project work were shared with teammates.

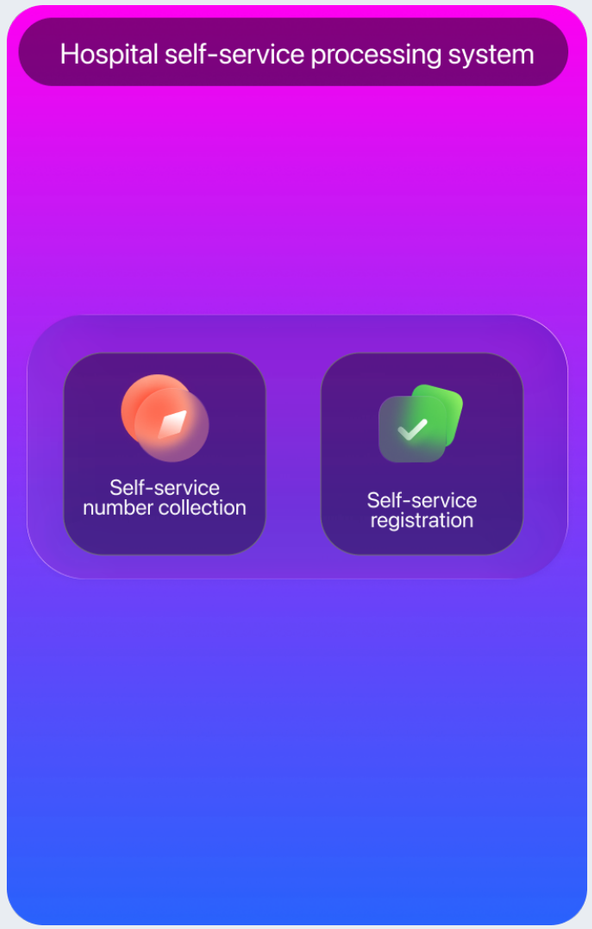




Original Figma exports • academic prototype, not a clinical service

A VISUAL SYSTEM ACROSS TOUCHPOINTS

One journey needs one visual language.



Original kiosk concept / Figma node 353:3581

OBSERVATION FROM THE ARTEFACTS

Mobile and kiosk share expressive icons, but use different surface and colour treatments.

TEACHER FEEDBACK TO CARRY FORWARD

The final assessment recognised promising mobile/tablet interfaces and a refined poster style. It also called for stronger hierarchy, kiosk consistency and accessibility justification.

DESIGN CONTRIBUTION

My Figma work explored a visual language across mobile and kiosk touchpoints. I would carry the assessment feedback into clearer hierarchy, stronger cross-touchpoint consistency and accessibility decisions.

Academic team concept / original Figma interface exports.